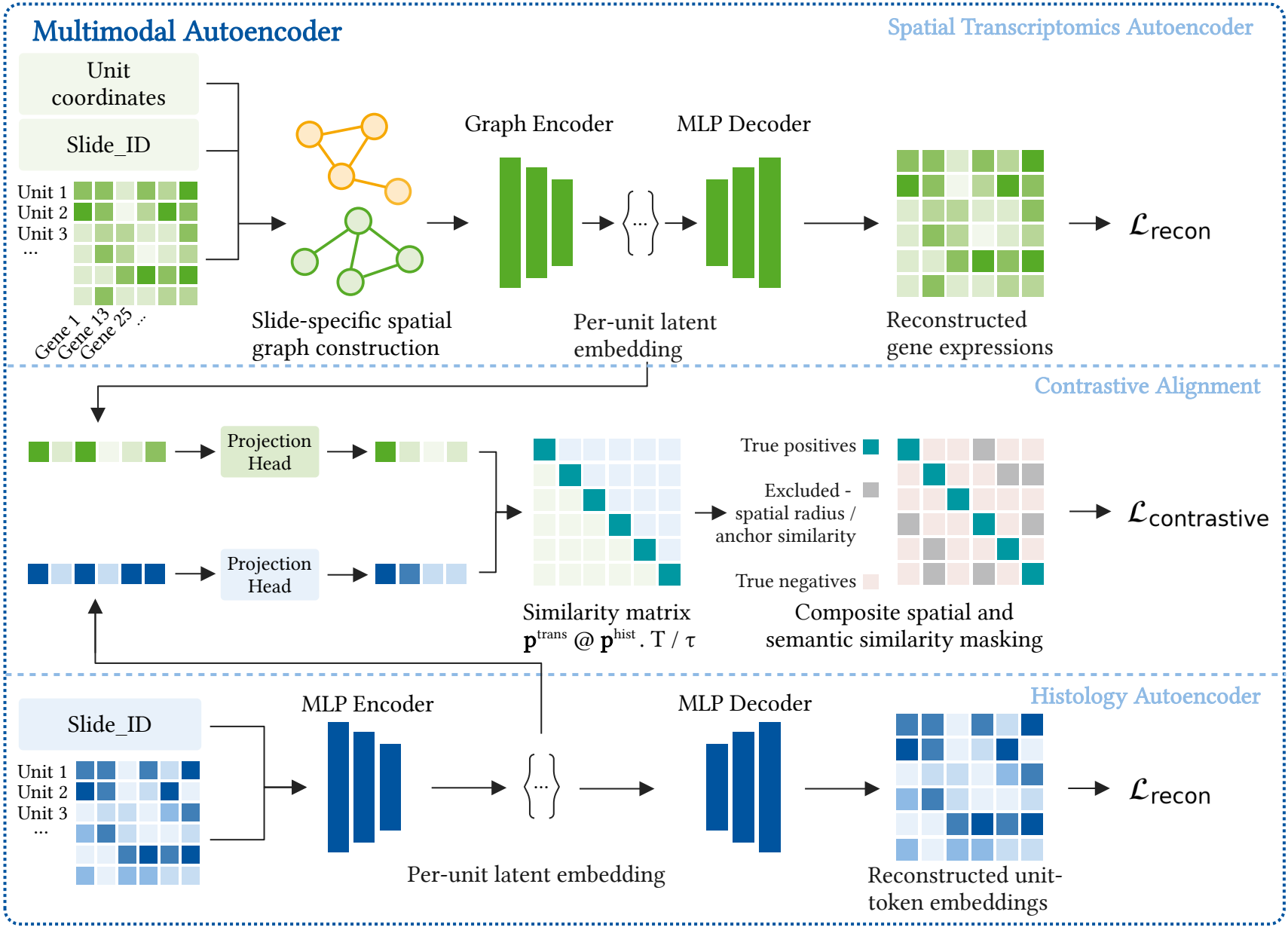
Multimodal Autoencoder
Spatial Transcriptomics Autoencoder
Unit coordinates
Slide_ID
Unit 1
Unit 2
Unit 3
...
Gene 1
Gene 13
Gene 25
...
Slide-specific spatial graph construction
Graph Encoder
MLP Decoder
Per-unit latent embedding
Reconstructed gene expressions
$\mathcal{L}_{recon}$
Contrastive Alignment
Projection Head
Projection Head
Similarity matrix
$\mathbf{p}^{trans} @ \mathbf{p}^{hist} . T / \tau$
True positives
Excluded - spatial radius / anchor similarity
True negatives
Composite spatial and semantic similarity masking
$\mathcal{L}_{contrastive}$
Histology Autoencoder
Slide_ID
Unit 1
Unit 2
Unit 3
...
MLP Encoder
Per-unit latent embedding
MLP Decoder
Reconstructed unit-token embeddings
$\mathcal{L}_{recon}$